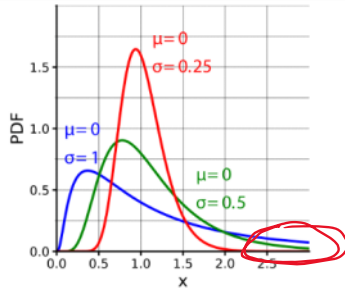


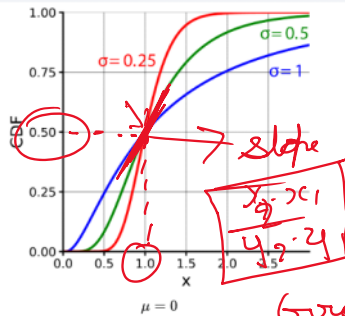
### Log-normal distribution

#### Probability density function



Identical parameter  $\mu$  but differing parameters  $\sigma$

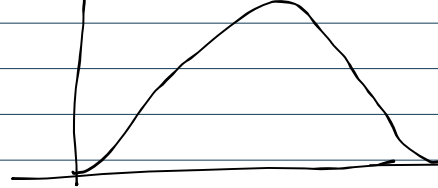
#### Cumulative distribution function



In probability theory, a **log-normal** (or **lognormal**) distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed. Thus, if the random variable  $X$  is log-normally distributed, then  $Y = \ln X$  has a normal distribution. Equivalently, if  $Y$  has a normal distribution, then the exponential function of  $Y$ ,  $X = \exp(Y)$ , has a log-normal distribution. A random variable which is log-normally distributed takes only positive real values.

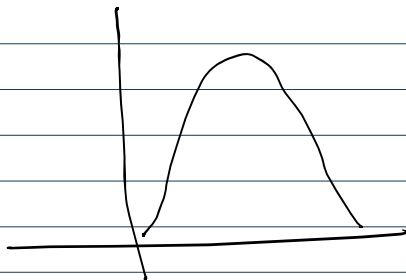
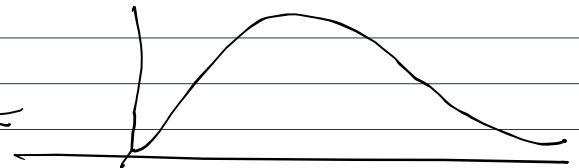
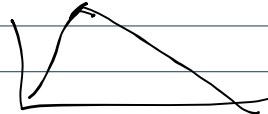
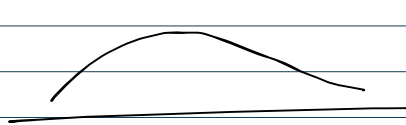
$$x = \log \text{Normal}(\mu, \sigma)$$

Right Skew distribution

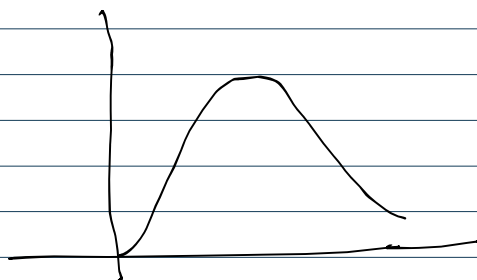
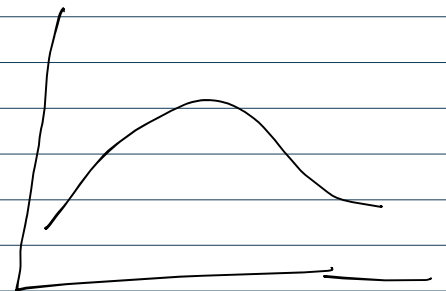


$$y = \ln(x)$$

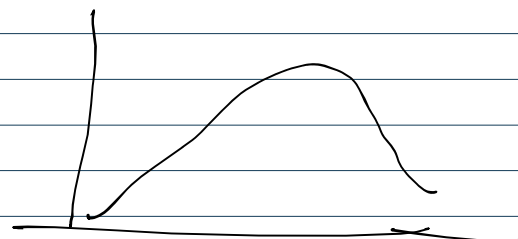
Natural log



$$x = \ln(y)$$



$$y = \ln(x)$$



- Ex: -
- ① width distribution of the users
  - ② length of the files
  - ③ online time for artists

